

# Software Engineering Insights 2025: AI, Trends, and Industry Challenges

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# Summary

This is a comprehensive analysis of the current landscape and future directions in the software engineering field, emphasizing the integration of artificial intelligence (AI) and the evolving demands of the industry. Compiled through a synthesis of peer-reviewed literature and extensive surveys, the report reflects significant trends, challenges, and innovations impacting both the education and practice of software engineering. With a global market projected to grow from USD 47.3 billion in 2022 to USD 89.3 billion by 2030, this report serves as a crucial resource for stakeholders aiming to navigate the complexities of modern software development.[\[1\]](#) [\[2\]](#)

Notably, the report highlights the rapid adoption of AI tools, such as ChatGPT and CoPilot, which are reshaping educational methodologies and professional practices. While these technologies promise enhanced productivity and learning outcomes, they also raise ethical concerns related to plagiarism and misuse, prompting discussions on the balance between leveraging AI and ensuring academic integrity.[\[1\]](#) [\[3\]](#) The findings underscore a dichotomy in user satisfaction with collaborative tools that facilitate human interaction while revealing significant challenges, such as the skills gap exacerbated by the "Great Resignation" and the ongoing impact of the COVID-19 pandemic on the workforce.[\[2\]](#) [\[4\]](#)

The report identifies critical emerging trends, including the dominance of cloud computing, advancements in digital transformation, and the increasing reliance on AI-driven solutions. These trends not only highlight opportunities for innovation but also emphasize the necessity for organizations to address integration and collaboration challenges within their software development processes. Additionally, the report underscores the importance of fostering industry-education collaborations to better prepare future professionals for the complexities of a rapidly evolving tech landscape.[\[5\]](#) [\[6\]](#)

As a reflection of the current state of the industry, this report builds on previous iterations by incorporating more refined methodologies and thematic analyses. This

evolution allows for a deeper understanding of the interdependencies between technology and human collaboration, ultimately guiding the software engineering community towards effective strategies for adaptation and growth in a competitive environment.[\[7\]](#) [\[8\]](#)

## Methodology

### Study Design

The methodology employed in this analysis comprises various approaches aimed at extracting and synthesizing insights from the reviewed literature. The primary goal is to investigate the use and efficacy of conversational agents (CAs) in software engineering education and industry settings. Various study methods were utilized across the papers reviewed, including experimentation with conversational agents, user studies involving questionnaires, interviews, and controlled prototype experiments.[\[1\]](#)

### Corpus Analysis

#### Publication Years and Countries

The corpus of papers analyzed spans publications from 2018 to January 2024. A marked increase in the volume of publications was observed, particularly from 2018 to 2023, across three main categories: Software Engineering Industry (SE\_I), Computer Science Higher Education (CS\_HE), and Software Engineering in Higher Education (SE\_HE). Notably, the increase was most significant in the SE\_I category, while SE\_HE publications remained comparatively limited.[\[1\]](#) The analysis anticipates a continuation of this trend into 2024.

#### Thematic Mapping and Reflexive Approach

A detailed thematic analysis was conducted using Reflexive Thematic Analysis (RTA) combined with Large Language Model (LLM)-based tools and a manual review process

to ensure the reliability of findings. This reflexive approach allowed for the re-evaluation of assumptions and interpretations, ensuring that identified themes genuinely reflected the data and were pertinent to the research questions posed.[\[1\]](#)

## Methodology Categorization

For the SE\_HE category, the full texts of the papers were reviewed, and the research was categorized according to the Software Development Lifecycle and the Guide to the Software Engineering Body of Knowledge (SWEBOK). This categorization focused on capturing main research questions, methodologies employed, and findings based on relevant criteria outlined in the study.[\[1\]](#)

## Data Collection and Analysis

The research utilized a rapid review approach guided by the PRISMA framework, employing a systematic selection process to include studies in the review. The search strategy encompassed key databases such as IEEE Explorer, ACM Digital Library, and ScienceDirect, with Google Scholar serving as an additional resource for capturing recent contributions published in arXiv. Only peer-reviewed articles were included, with the exception of one open-source tool that was relevant to the study.[\[1\]](#)

## Evaluation of Conversational AI

The analysis incorporated various studies assessing the capabilities of generative AI tools like ChatGPT and CoPilot in educational contexts. This included user studies involving surveys and observational methods to evaluate how these tools can support student learning and instructional needs, including preparing course materials and grading.[\[1\]](#) The findings indicate a growing curiosity regarding the effectiveness of these tools in programming education and other subjects, despite noted concerns about their potential for misuse and plagiarism.[\[1\]](#)

# Key Findings

## Thematic Analysis and Research Clusters

A detailed thematic analysis was conducted to derive insights from the existing literature on software engineering, specifically through the lens of human-computer interaction and collaboration tools in the development process. The analysis utilized Reflexive Thematic Analysis (RTA) in conjunction with Large Language Model (LLM)-based tools, ensuring a reliable examination of abstracts and full papers related to Software Engineering (SE) practices.[\[1\]](#) This approach facilitated the identification of significant themes that reflect the core concerns and challenges in the field.

## User Satisfaction with Collaborative Tools

Surveys indicated a high level of user satisfaction with virtual assistants designed to facilitate human collaboration in software development. These tools were effective in addressing common challenges associated with finding and collaborating with human partners, thereby improving overall productivity and workflow efficiency.[\[1\]](#) This insight is crucial as organizations increasingly rely on technology to streamline their operations and enhance team collaboration.

## Co-Occurrence Analysis of Research Topics

A term co-occurrence analysis was performed to uncover clusters of research topics and their interconnections within the software engineering domain. The resulting network graph illustrated the relationships and frequencies of various terms extracted from author keywords. This visualization highlighted key terms that emerged frequently, suggesting areas of high research interest and potential future exploration within the field.[\[1\]](#)

# Challenges Facing Software Engineering

As the software engineering landscape continues to evolve, several external and market-driven challenges have emerged. These include rapid technological advancements, skill shortages exacerbated by the "Great Resignation," and the increasing demand for quick software delivery without compromising quality. Companies are under pressure to innovate while navigating potential talent shortages, which complicates the recruitment and retention of skilled professionals.[\[2\]](#) [\[3\]](#) The ongoing impact of the COVID-19 pandemic further complicates the landscape, making it essential for organizations to adapt their operational strategies to remote and hybrid work environments.

## Financial Growth and Market Trends

The software engineering market is projected to experience substantial growth, with the global market size expected to rise from USD 47.3 billion in 2022 to USD 89.3 billion by 2030, reflecting a compound annual growth rate (CAGR) of 11.8%.[\[2\]](#) This growth is indicative of the increasing reliance on software solutions across various industries, highlighting the importance of software engineering in modern business operations.

## Future Directions

The insights derived from this report emphasize the necessity for software engineers to balance technological capabilities with human collaboration to maximize potential outcomes. As the industry anticipates exciting innovations, understanding the interdependencies between different research areas and the underlying challenges will be vital for steering future developments in software engineering.[\[1\]](#)

## Emerging Trends

The landscape of software engineering is undergoing significant transformations as organizations adapt to evolving technologies and increasing demands for digital

solutions. In 2024, several key trends are emerging that shape the future of software development and deployment.

## Artificial Intelligence in Software Development

Artificial Intelligence (AI) has become a cornerstone of modern software engineering, influencing various aspects from automated code reviews to predictive algorithms.[\[4\]](#) AI-powered tools are now integral for developers, assisting in tasks such as code suggestion, auto-completion, and bug detection. These tools leverage historical data and vast codebases to enhance code quality and accelerate development cycles, making AI not just a buzzword but a fundamental component of the software development process.[\[5\]](#) [\[6\]](#)

Moreover, as companies strive to keep pace with the demand for AI-driven services, it is projected that businesses will utilize no-code project management and AI development tools for a significant portion of their automation initiatives by the end of 2024.[\[4\]](#) This trend illustrates a shift towards simplifying development processes and increasing efficiency.

## Cloud Computing Dominance

Cloud computing continues to reinforce its position as a dominant force in the industry, with an emphasis on serverless computing and microservices. This shift allows organizations to leverage IT resources on-demand over the internet without managing underlying infrastructures directly.[\[4\]](#) The consolidation of software tools into all-in-one platforms is also gaining traction as companies seek to enhance security and reduce vulnerabilities associated with using multiple systems.[\[4\]](#)

## Digital Transformation in Education

Higher education institutions are increasingly focusing on digital transformation and resilience. As part of this effort, they are investing in technology that meets diversity, equity, and inclusion (DEI) standards while ensuring accessibility for all users.[\[7\]](#) The

move towards online and hybrid learning models has prompted colleges and universities to implement new programs that cater to diverse learning needs, thus fostering an inclusive educational environment.[\[7\]](#)

## Advanced Connectivity and Emerging Technologies

The advancement of technologies such as advanced connectivity, generative AI, and cloud and edge computing has entered the scaling stage of adoption, indicating a robust integration within various industries.[\[8\]](#) These technologies are serving as enablers for the adoption of other emerging trends, driving innovation and efficiency in software development.

## Challenges and Future Directions

Despite the rapid pace of innovation, organizations face challenges such as rising cybersecurity threats and the complexities of integrating AI within DevOps practices.[\[5\]](#) [\[6\]](#) Striking a balance between AI-driven development and maintaining effective operational processes will be a critical focus for software engineering teams moving forward.

## Industry Challenges

### Economic Pressures and Growth Limitations

The 2023 report highlighted a significant economic downturn that impacted the growth trajectory of many technology companies, particularly in the data sector. Despite a rise in market indexes, this growth was largely driven by a select number of firms at the forefront of the Generative AI (GenAI) revolution, primarily those specializing in hardware and foundation models. The remainder of the tech market continued to struggle, prompting companies to pivot and expand their product offerings into adjacent domains in a bid to boost revenue streams.[\[9\]](#)



## The AI Skills Gap

As the demand for AI technologies surged, the industry faced a critical skills gap, limiting the ability of organizations to implement AI effectively. This shortage led to a rush in adopting AI solutions without a comprehensive understanding of the technology, often resulting in suboptimal or misaligned products. Consequently, businesses found themselves needing to strategize around hiring and training skilled professionals to harness AI's full potential while maintaining a focus on innovation and growth.[\[10\]](#)

## Ethical and Regulatory Complexities

Navigating the ethical and regulatory landscape has become increasingly complex for technology companies. As innovations in AI, blockchain, and infrastructure continue to emerge, organizations must balance the pursuit of technological advancement with the need for responsible and sustainable practices. Moving forward, strategic investment in these areas will be essential for fostering an environment that encourages continuous learning while ensuring ethical use of technology.[\[10\]](#) [\[8\]](#)

## Cloud Spending and Financial Management

With the rise of cloud technologies, organizations have faced mounting pressures to manage cloud expenditures effectively. Runaway spending became a widespread issue as resources could be provisioned with a single click. While hyperscalers provided tools for better oversight, many of these financial operations (FinOps) tools still required manual inputs, resulting in limited visibility. The integration of AI into financial management is seen as a necessary evolution, offering real-time analysis and predictive capabilities that could optimize costs amidst rising cloud demands.[\[11\]](#) [\[12\]](#)

## Code Quality and Complexity

The increasing reliance on AI for software development raises concerns regarding code quality and complexity. Companies leveraging generative AI tools have reported

significant savings in development time and costs. However, this efficiency can come with risks, such as the generation of unnecessarily complex code that may lead to technical debt and hinder long-term performance. As organizations adopt these technologies, they must be vigilant about maintaining a balance between automation and the quality of the code produced.[\[13\]](#) [\[14\]](#)

## Integration and Collaboration Challenges

As continuous integration and continuous delivery (CI/CD) practices evolve, challenges remain in ensuring seamless collaboration across various tools and technologies within platform engineering. The integration of AI and machine learning into these workflows holds promise for increased efficiency, yet it also necessitates a robust strategy to manage the complexities that arise from integrating multiple systems and processes. The future of CI/CD will depend on fostering greater interoperability and standardization to enhance overall productivity within the software development lifecycle.[\[15\]](#) [\[16\]](#)

## Recommendations

### Embrace Personalized Learning Approaches

Educational institutions and companies should prioritize personalized learning to accommodate diverse learning styles and paces. This can be achieved through tailored lesson plans, assessments, and the integration of AI-driven tools that adjust learning materials in real-time based on individual progress. As the landscape evolves, experiments in personalized AI tutoring are expected to gain traction by 2025, providing dynamic and engaging learning experiences for students.[\[17\]](#)

### Foster Industry-Education Collaborations

To address the skills gap in high-demand areas such as artificial intelligence and cybersecurity, partnerships between educational institutions and industry leaders are essential. These collaborations can ensure that curricula are aligned with real-world

needs, enabling students to acquire relevant skills before entering the workforce.[\[17\]](#)  
[\[18\]](#) Institutions should proactively engage with employers to continuously update their programs, preparing graduates to meet industry demands effectively.

## Promote Continuous Learning and Upskilling

As technology advances rapidly, continuous learning must be ingrained in workplace culture. Companies should invest in employee training programs that focus on emerging technologies and soft skills. Offering workshops, courses, and access to online learning platforms will help employees adapt and thrive in changing environments. Notably, companies like IBM are committing to extensive training initiatives, highlighting the importance of workforce development in today's job market.[\[18\]](#) [\[19\]](#)

## Invest in AI Certifications

Individuals seeking to enhance their careers should consider obtaining AI certifications that match their skill levels and career goals. These certifications can range from beginner-friendly courses to advanced programs aimed at experienced professionals. Many organizations offer free or low-cost options, making it feasible for anyone to start their journey into AI without significant financial barriers.[\[20\]](#) [\[18\]](#) Furthermore, obtaining certifications can lead to better job prospects and potentially higher salaries.[\[19\]](#)

## Adapt to Hybrid Work Environments

As many organizations navigate the shift to hybrid and remote work arrangements, it is crucial for institutions to create flexible work policies that attract and retain talent. Offering remote work options can broaden the recruitment pool and help address staffing shortages in critical areas. Institutions should actively promote their hybrid models and adapt their operations to support a diverse workforce, which can lead to improved employee satisfaction and productivity.[\[7\]](#)

By implementing these recommendations, educational institutions and companies can better prepare themselves for the evolving landscape of software engineering and technology in 2025 and beyond.

## Comparison with Previous Reports

The **State of Software Engineering Insights 2024** Report builds upon methodologies and findings from prior editions, incorporating data from six Developer Nation surveys conducted by SlashData between Q3 2020 and Q1 2023, covering insights from over 125,000 respondents worldwide.[\[21\]](#)

Past reports focused on key advancements in CI/CD, software quality, and AI adoption. For instance, the **World Quality Report**, published annually since 2009, serves as a benchmark for trends in application quality and software testing. The 2024 edition gathered insights from over 1,750 senior executives across industries, highlighting the increasing importance of quality assurance in modern software development.[\[22\]](#)

This year's report refines its thematic analysis, categorizing research into **Software Engineering in Higher Education (SE\_HE)**, **Computer Science in Higher Education (CS\_HE)**, and **Software Engineering Innovations (SE\_I)**. This methodological shift allows for a deeper, more structured exploration of evolving trends, ensuring relevance in today's rapidly changing software engineering landscape.[\[23\]](#)

## Conclusion

The State of Software Engineering Insights 2025 report highlights the transformative impact of AI, cloud computing, and DevOps on modern software development. Organizations must address industry challenges, including talent shortages, ethical AI implementation, and financial pressures associated with cloud expansion. By fostering industry-academia collaborations, investing in workforce upskilling, and embracing AI-driven automation, companies can stay ahead in an increasingly competitive landscape. Future success in software engineering will depend on a balance between

technological innovation and strategic workforce development. As software engineering continues to evolve, embracing adaptability and continuous learning will be key to navigating its future.

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